Lab 3, Act 2

Counting detections: binomial or Poisson?

NRES 746

Answer key. Answers are in magenta.

Picking up where we left off

In Act 1 you dealt with *uncertainty about a single flagged image*. Now set that aside – assume genetic testing has confirmed the target carnivore is present at a particular site (no more classifier doubt to worry about). The new question is about *detection*: even when the animal is really there, a camera doesn't catch it every single night. Suppose, based on many past deployments at similar sites, biologists estimate that on any given night there's a $p = 0.3$ probability the camera captures at least one image of the animal, independent from night to night (assuming it is genuinely there!).

Part 1: Binomial by hand (individual, ~15 min)

A camera is deployed for $N = 8$ nights.

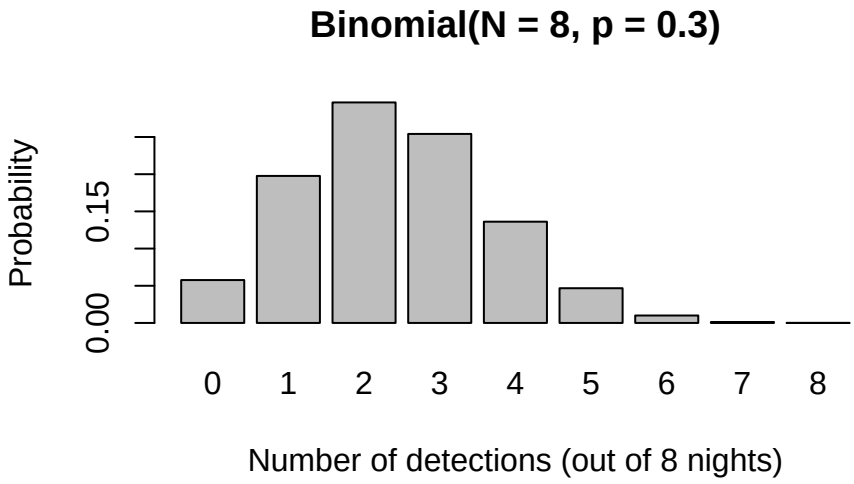
1a. Why is "number of nights with a detection, out of 8" a reasonable binomial random variable here? Name the assumptions the binomial distribution requires, and say briefly why each one seems plausible (or not) for this camera. Feel free to refer to the Bolker book here.

Fixed number of trials (8 nights, by design); two outcomes per night (detect or not); independent nights (plausible, but storms, malfunctions, or an animal revisiting a spot could link nights); constant p (questionable: moon, weather, or the animal getting used to the camera can change it).

1b. Using the binomial pmf, $P(X = k) = \binom{N}{k} p^k (1-p)^{N-k}$, compute by hand the probability of exactly 3 detections in the 8 nights ($N = 8$, $k = 3$, $p = 0.3$). Show your work.

$P(X = 3) = \binom{8}{3}(0.3)^3(0.7)^5 = 56 \times 0.027 \times 0.16807 \approx 0.2541$

Below is the full distribution of detections across all 9 possible outcomes ($k = 0$ to 8), for you to check your answer against and read off the most likely outcome.

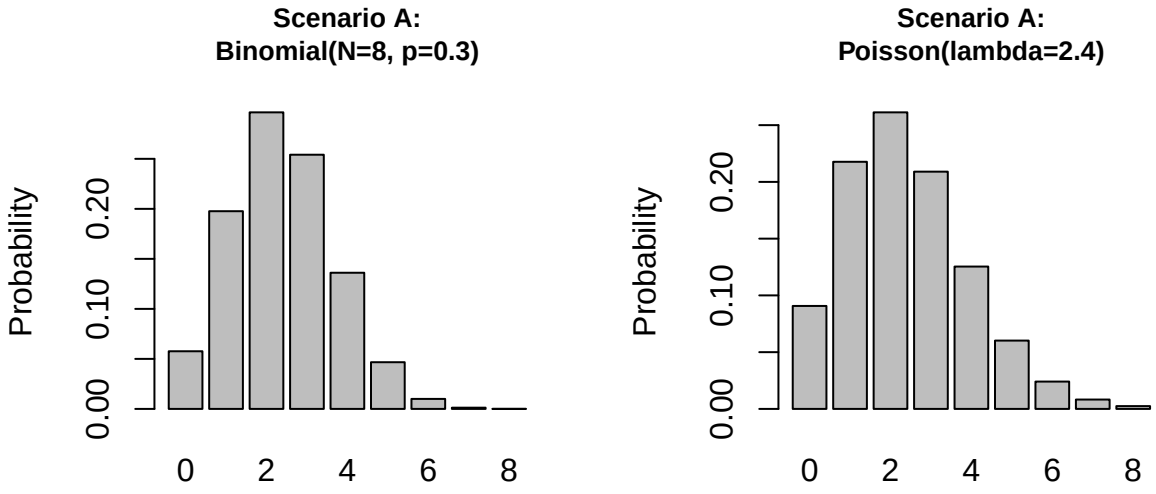


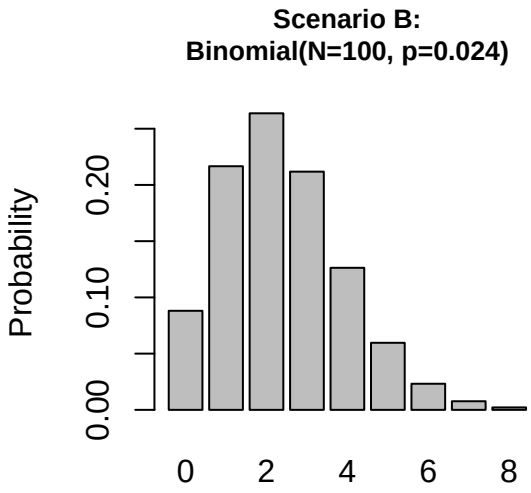
1c. The mean of this distribution is $Np = 8 \times 0.3 = 2.4$, but the tallest bar (the mode) sits at a different whole number. Why can't the mode land exactly on the mean here? And what does the direction of the gap between them (mode below the mean, or above?) tell you about the shape of this distribution?

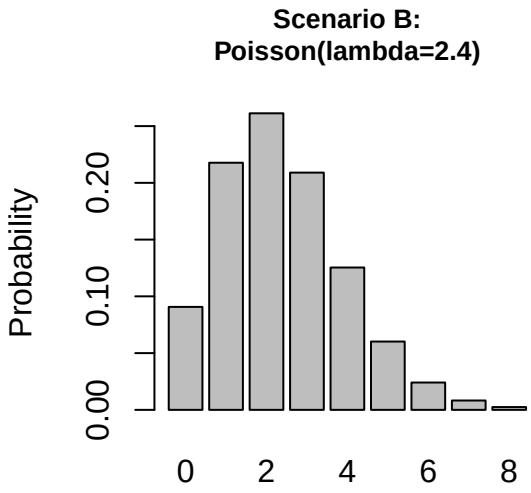
The mode is 2. Counts must be whole numbers, but the mean is an average and can fall between them. Mode below the mean means right skew: bounded at 0 with a longer upper tail pulling the mean up.

Part 2: Comparing to Poisson (individual, ~15-20 min)

A colleague suggests it would be simpler to just model detections with a Poisson distribution instead, using the same mean, $\lambda = 8 \times 0.3 = 2.4$. Below are two scenarios, each comparing a binomial distribution against a Poisson with the same mean.







2a. Look closely at both pairs of bar charts. In Scenario A, where specifically do the binomial and Poisson bars disagree most – the center, the spread, or the tails? Does Scenario B show the same kind of disagreement, a different kind, or none at all?

Scenario A: the spread. The binomial is taller in the middle (2 to 4); the Poisson puts more probability in both tails. Scenario B: almost no disagreement.

2b. Compute the variance of each distribution by hand, using $\text{Var} = Np(1-p)$ for the binomial and $\text{Var} = \lambda$ for the Poisson.

A binomial: $8 \times 0.3 \times 0.7 = 1.68$. B binomial: $100 \times 0.024 \times 0.976 = 2.3424$. Poisson (both): $\lambda = 2.4$.

2c. Does the pattern of variances you just computed explain what you saw by eye in 2a?

Yes: in A the Poisson variance (2.4) is about 40% larger than the binomial's (1.68), so it's wider; in B they're nearly equal (2.40 vs. 2.34).

2d. Finish this rule of thumb in your own words: *"A Poisson distribution is a good stand-in for a binomial distribution when. . . "*

...the number of trials is large and the probability per trial is small, so $1 - p \approx 1$ and $Np(1-p) \approx Np = \lambda$.

2e. Back in the original 8-night scenario ($p = 0.3$): name one realistic reason the constant, independent per-night detection probability assumption from 1a might be violated in the field.

E.g. a multi-night storm suppresses detections several nights in a row (breaks independence), or moon phase, season, or the animal learning to avoid the camera changes p.

Group discussion and revision (~5 min)

Compare your variances and rule of thumb from 2d with your group.

Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion ("muddiest points") or topics you want to review or dust off going forward.

Open-ended.